



Modelling ammonia volatilization

SEVEN CROPPING SYSTEMS OF THE TRAJECTOIRE
EXPERIMENTAL PLATFORM (2018–2025)

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Background

 94%

OF ATMOSPHERIC NH_3
FROM AGRICULTURE

 > 50%

OF $\text{PM}_{2.5}$ MASS
FORMED FROM NH_3

 ×273

GWP_{100} OF N_2O
(INDIRECT NH_3 PATHWAY)



SEVEN SYSTEMS

SCA · GHG · HYP · CTR · LI · OF · MF

152 EVENTS OVER TWO ROTATIONS (2018–2025)

Contrasted objectives: yield, GHG, low-input, organic, mixed

THE TRANSITION

CTR → COM CROP-ORIENTED METHANIZATION
MF → LOM LIVESTOCK-ORIENTED
METHANIZATION

Digestate replaces conventional inputs in rotation 2

The Trajectoire platform

GRIGNON, ÎLE-DE-FRANCE

KEY FIGURES

10 ha
Experimental
platform surface

7

5 yr

2 000
Field
measurements per
year

SEVEN CONTRASTING SYSTEMS

Each strip $\approx$ 1.4 ha · same soil (silt-loam, pH 7.66) · same climate

SCA	SOIL CONSERVATION AGRICULTURE	No tillage · permanent soil cover
GHG	LOW CARBON	Nitrification inhibitors · low fossil fuel use
HYP	HIGH YIELD PERFORMANCE	Maximize yield · conventional practices
CTR	CONTROL	HVE-3 certified reference practices
LI	LOW INPUT	Reduce fertilizer & plant-protection inputs
OF	ORGANIC FARMING	Organic inputs only · mechanical weeding
MF	MIXED FARMING	Livestock effluents · slurry, solid manure



COMMON 5-YEAR ROTATION:

wheat → second cereal → rapeseed (faba bean in OF) → wheat → maize · R1 (2017–2022) → R2 (2022–2027, in progress)

Source: AgroParisTech (2024). Trajectoire · Review of the first crop rotation 2017–2022.

Research question & objectives



HOW MUCH AMMONIA IS LOST ACROSS THE SEVEN CROPPING SYSTEMS OF TRAJECTOIRE, AND HOW MUCH DOES IT CHANGE AFTER THE TRANSITION TO ANAEROBIC DIGESTION?

Objectives

- 1 QUANTIFY NH_3 -N EMISSIONS FROM ALL 152 FERTILIZATION EVENTS ACROSS THE SEVEN CROPPING SYSTEMS
- 2 MEASURE THE CHANGE IN EMISSIONS AFTER TWO SYSTEMS TRANSITION TO ANAEROBIC DIGESTION
- 3 TRANSLATE VOLATILIZED N INTO AN INDIRECT- N_2O CLIMATE COST (IPCC 2019 METHODOLOGY)

Methods

Emission-factor model



EMEP Tier 2

HOW IT WORKS

$\text{EMISSION} = N \text{ APPLIED} \times \text{EF}(\text{PRODUCT}, \text{PH})$

STRENGTHS

- ✓ COMPLETE INVENTORY (ALL 152 EVENTS)
- ✓ SIMPLE, REPRODUCIBLE, COMPARABLE
- ✓ POLICY-STANDARD METHOD

LIMITS

- ✗ FIXED FACTOR PER PRODUCT
- ✗ NO WEATHER SENSITIVITY

Process-oriented dynamic model



ALFAM2

HOW IT WORKS

FIRST-ORDER TRANSFER FROM TAN POOLS,
HOURLY

STRENGTHS

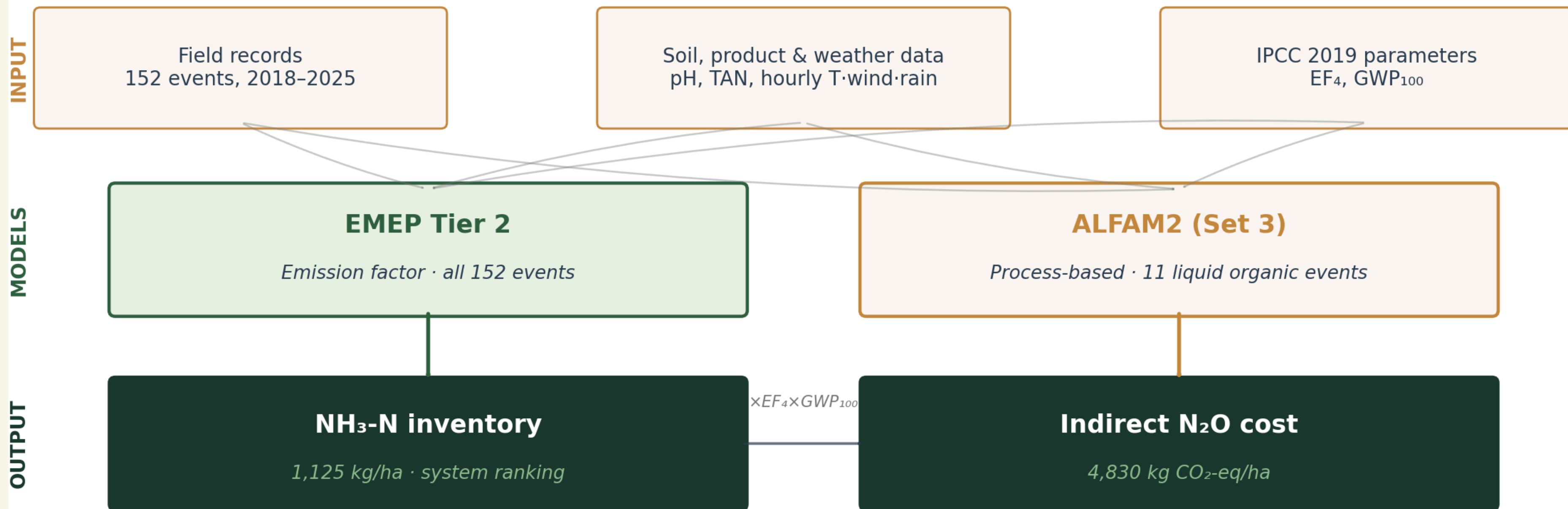
- ✓ RESOLVES WEATHER (T, WIND, RAIN)
- ✓ DIFFERENTIATES SLURRY VS DIGESTATE
- ✓ CALIBRATED ON >700 FIELD PLOTS

LIMITS

- ✗ ONLY LIQUID ORGANIC (11 EVENTS)
- ✗ EXTRAPOLATION AT CALIBRATION EDGES

WORKFLOW

From field data to indirect climate cost



Result 1 — EMEP inventory

1,125.5

KG NH₃-N HA⁻¹

Full inventory (2018–2025)

667.2

R1 · 87 events

458.3

R2 · 65 events

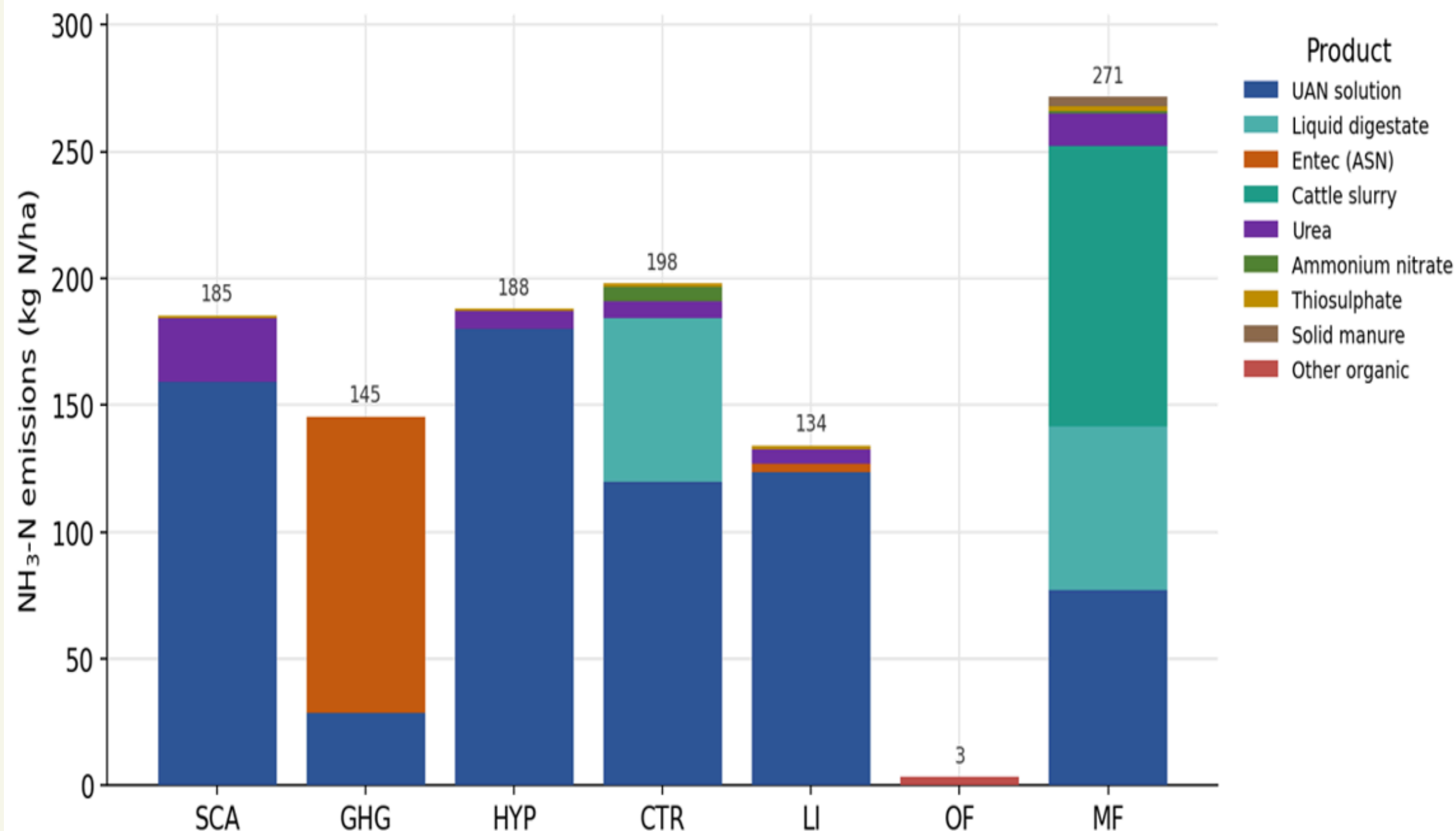
TOP INTENSITIES

MF 16.1% · CTR 14.7%

GHG 9.6% · OF 1.0%

NH₃-N emissions by system, stacked by product category

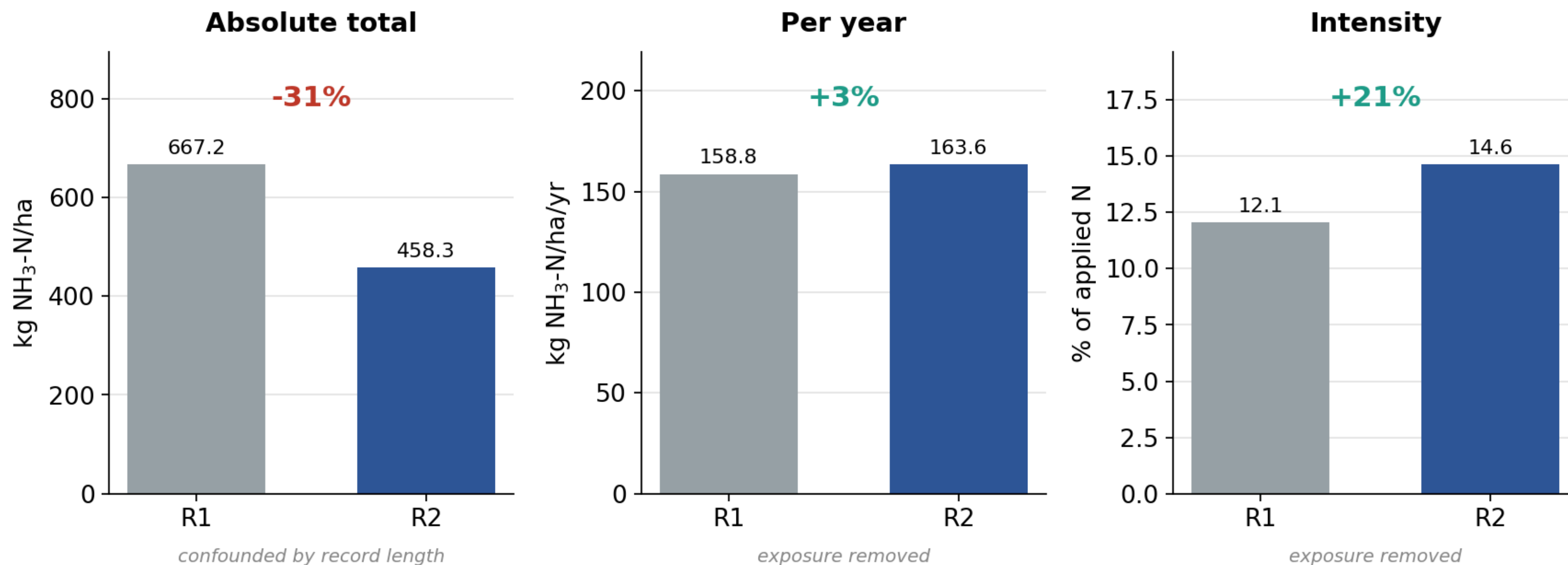
EMEP/EEA Guidebook 2023, Tier 2 — full inventory (2018-2025) · 1125 kg NH₃-N/ha across 152 events



RESULT 2 — THE METHANIZATION PARADOX

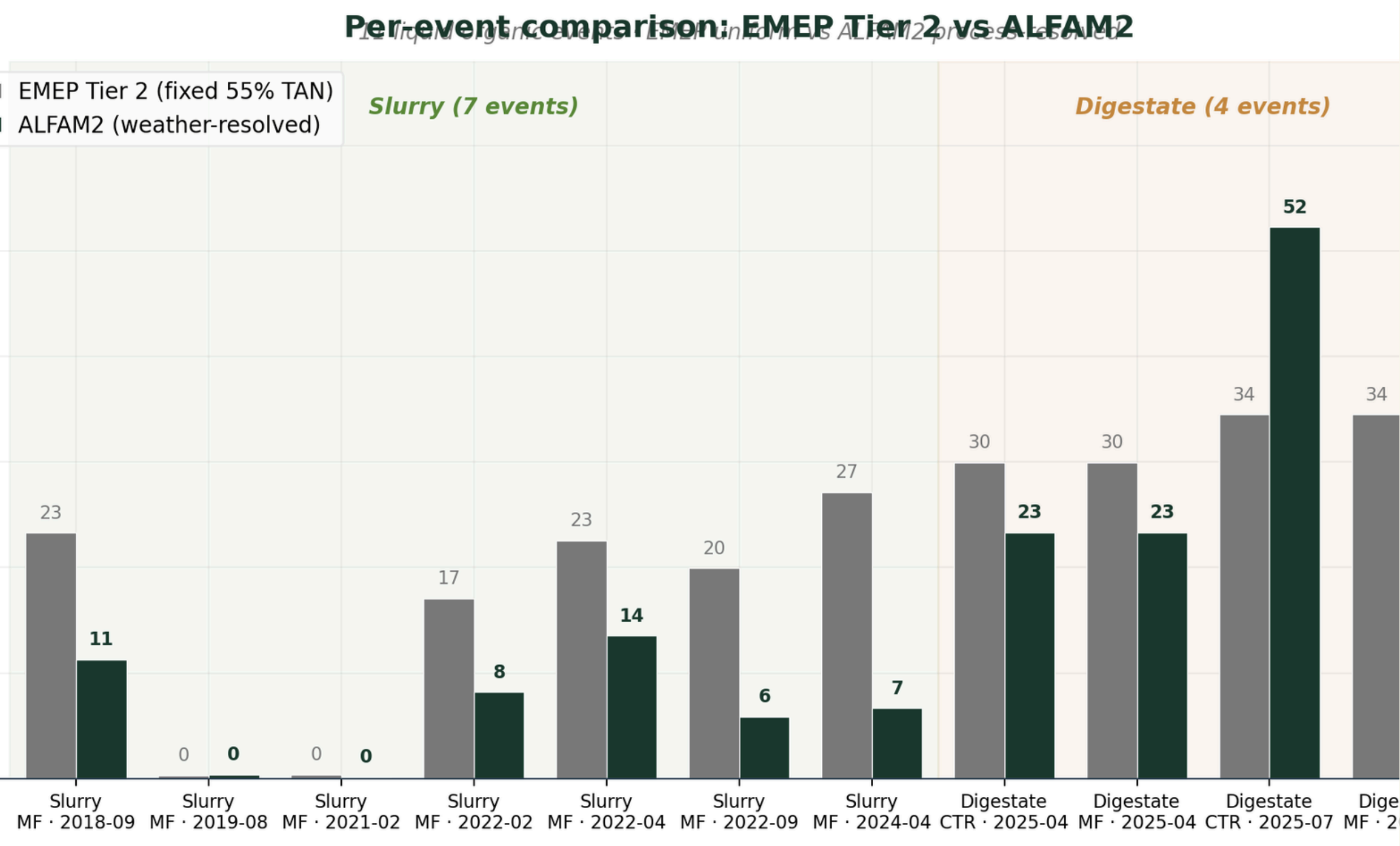
Comparing the two rotations: the absolute drop is a record-length effect

R1: 87 events, 4.2 yr, 5533 kg N/ha applied R2: 65 events, 2.8 yr, 3130 kg N/ha applied



Result 5 — EMEP vs ALFAM2 per event

Where the fixed factor over- or undershoots



SLURRY EVENTS ALFAM2 < EMEP

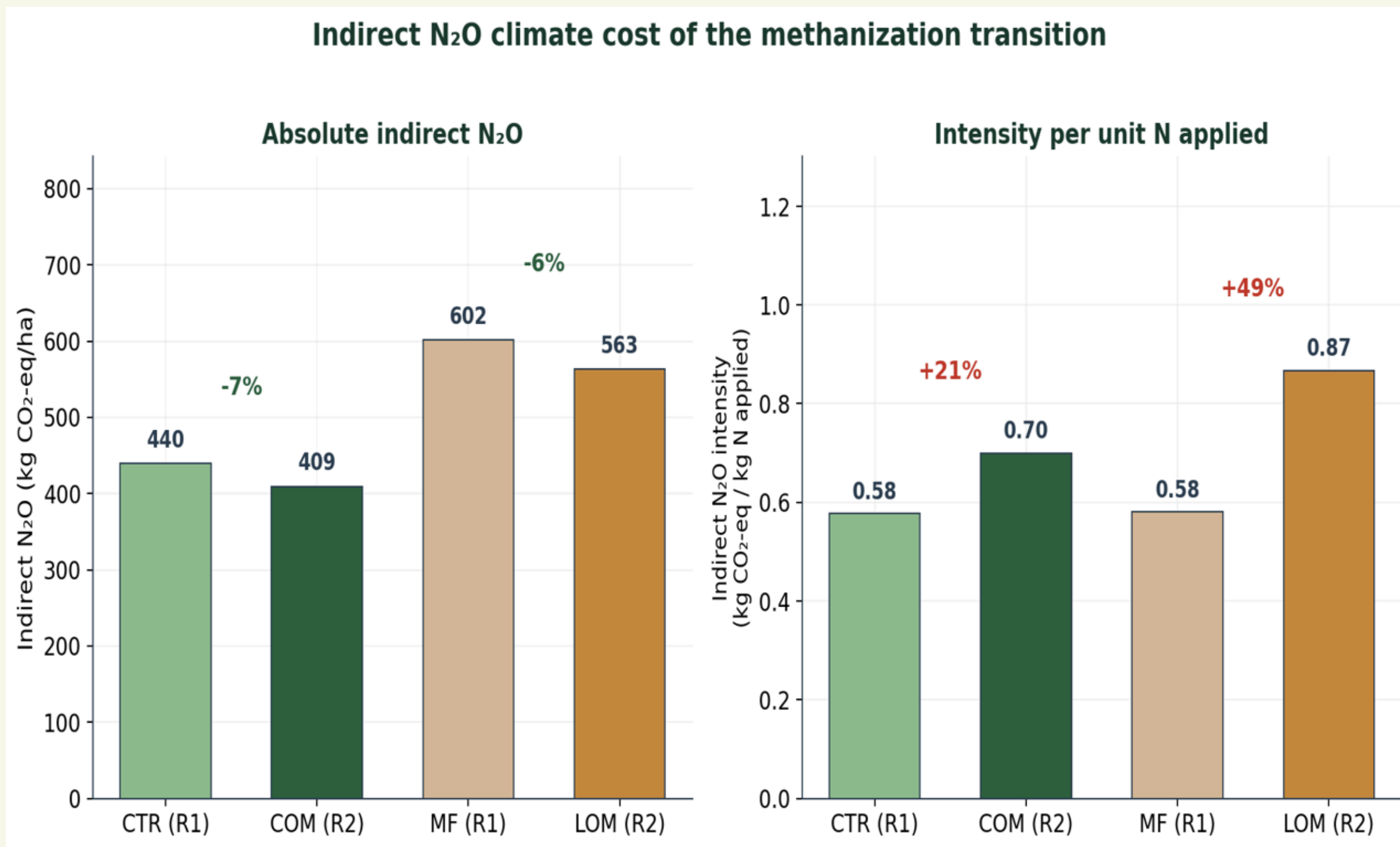
EMEP's 55% factor overestimates slurry, which volatilizes less due to lower pH and TAN

DIGESTATE EVENTS ALFAM2 > EMEP (SUMMER)

For warm-season digestate, ALFAM2 predicts ~52 kg N/ha versus EMEP's 34 — the fixed factor underestimates the actual risk

Result 6 — Indirect N₂O climate cost

NH₃ loss returns as N₂O · closing the nitrogen cascade



4,830

kg CO₂-eq ha⁻¹

platform total indirect N₂O

range: 966 – 8,691 (EF₄ uncertainty)

THE CHAIN

$$\text{N}_2\text{O} = \text{NH}_3\text{-N} \times \text{EF}_4 \times 44/28 \times \text{GWP}_{100}$$

EF₄ = 0.010 (IPCC 2019)

GWP₁₀₀ = 273 (AR6)

TRANSITION INTENSITY:

CTR→COM: +21%

MF→LOM: +49%

Conclusions

1

Complete inventory
1,125.5 kg NH₃-N/ha

2

Transition raises intensity
+21% (COM), +49% (LOM)

3

Digestate drives risk
~2× slurry, same
technique

4

Climate cost is real
~4,830 kg CO₂-eq/ha



PERSPECTIVES

PRE-TREATMENT

Acidification of digestate reduces pH
and TAN, cutting volatilization at
source

Fangueiro et al. (2015)

APPLICATION TIMING

Prefer cool seasons — spring/autumn
instead of summer to reduce warm-
weather peaks

Sommer & Hutchings (2001)

DIRECT MEASUREMENT

Validate high-risk events (summer
digestate) with field NH₃
measurements

Hafner et al. (2025)

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Thank you



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